Waiter

LongMemEval-S retrieval report

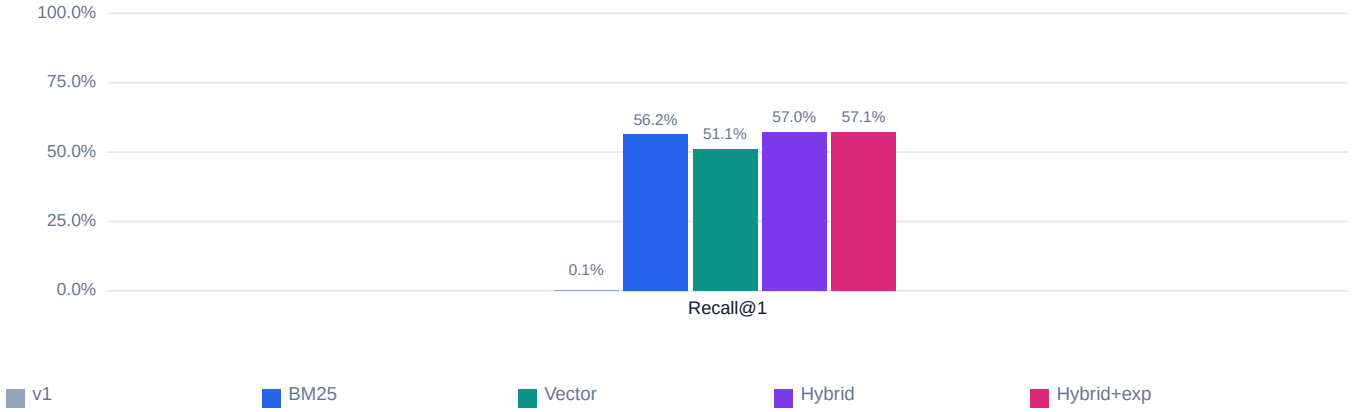
365 held-out questions · all-MiniLM-L6-v2 embeddings · local CPU only · no LLM answerer or judge

Generated 2026-10-27 · dataset: xiaowu0162/longmemeval-cleaned (S split)

Waiter results are retrieval-only; Supermemory figures are vendor-published and not a matched head-to-head run.

Headline

57.0% Recall@1 for Hybrid RRF — up from **0.1%** with the v1 keyword retriever.



Every question ships with ~1,22,373 tokens of history. Hybrid RRF puts an evidence session at rank 1 for 57.0% of questions while reading 15,904 tokens into a prompt — 13% of the history.

Verdict

Retrieval went from broken to usable: Recall@1 0.1% to 57.0%, MRR@20 0.3% to 93.0%. Recall@20 is already 97.8% with plain BM25, because this split only has ~48 sessions per question; it has little power to distinguish systems here. The honest read: Waiter now retrieves well for a lexical-first system, embeddings help most on preference questions, and answer quality was not measured. This report does not show that Waiter beats Supermemory.

1. What was measured, and what was not

LongMemEval has two halves. The retrieval half asks whether the right sessions come back for a question. The QA half feeds those sessions to a model and has a judge score the answer. This report covers the first half only.

Component	Status here	Why
Retrieval (Recall@k, NDCG, MRR)	Measured, this report	Runs entirely locally on CPU. No credentials needed.
QA accuracy (LLM answer + judge)	Not run	Needs an answerer model and an LLM judge with API access. Neither is available here, so no accuracy number is claimed.
Token cost	Measured	Character counts of retrieved sessions / 4, against full history.

Benchmark configuration

- Dataset: xiaowu0162/longmemeval-cleaned, split longmemeval_s_cleaned.json — 500 questions, 23,867 haystack sessions, ~1,22,373 tokens of history per question.
- Corpus per question: one memory per haystack session, scoped by userId = question_id. LongMemEval labels evidence at session level, so the session is the retrieval unit.
- Relevance: a retrieved session counts only if its id is in answer_session_ids.
- Split: 70/30 by question-id hash. 135 dev questions tuned the recency weight; the 365 test questions are reported and were never used for tuning.
- Embeddings: Xenova/all-MiniLM-L6-v2 (384-dim, quantised, mean-pooled, L2-normalised) on local CPU. Each session is embedded as a digest of its user turns, capped at 1,100 characters, because MiniLM reads 256 tokens and a session is ~2.5k.
- BM25 indexes the full session transcript, so the lexical side sees text the dense side cannot. That asymmetry is a property of the models, not a tuning choice, and it matters for the assistant-turn questions below.

Metrics

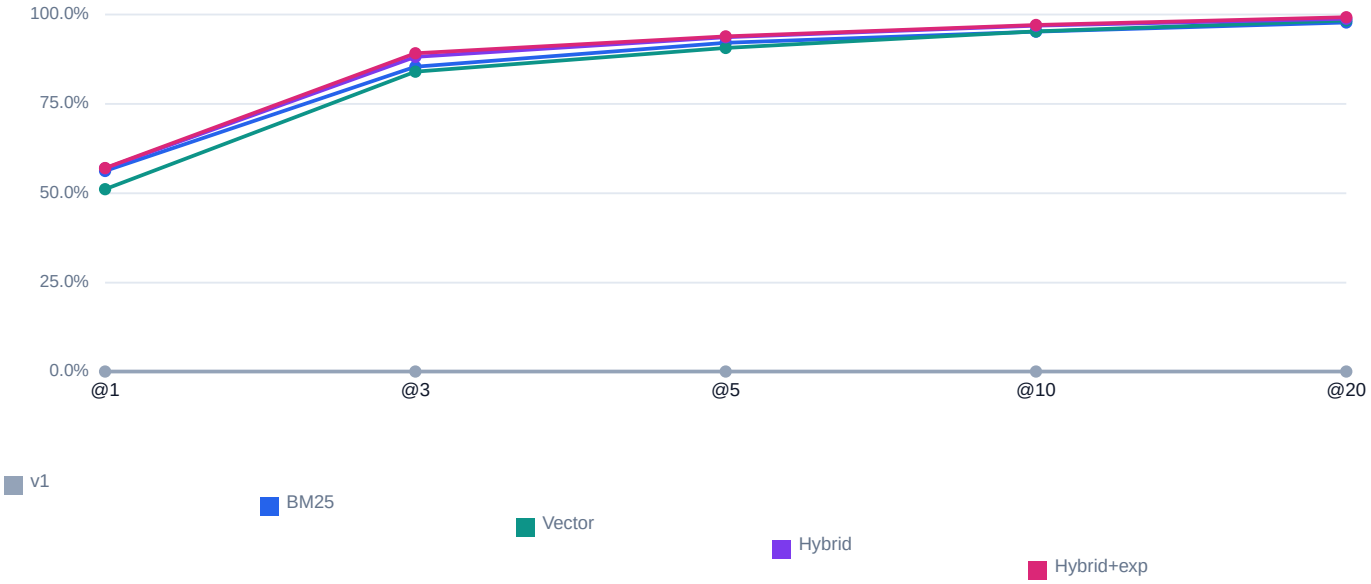
Metric	Definition
Recall@k	Fraction of a question's evidence sessions present in the top k results.
NDCG@10	Rank-aware gain, binary relevance, normalised by min(|evidence|, 10).
MRR@20	Reciprocal rank of the first evidence session found in the top 20.
Latency	Warm in-process query time; the BM25 index is built before timing starts.
± figures	95% confidence intervals across questions.

2. Retrieval results

Five retrievers, one corpus, 365 questions, identical labels. Only ranking differs, so the deltas are attributable to retrieval.

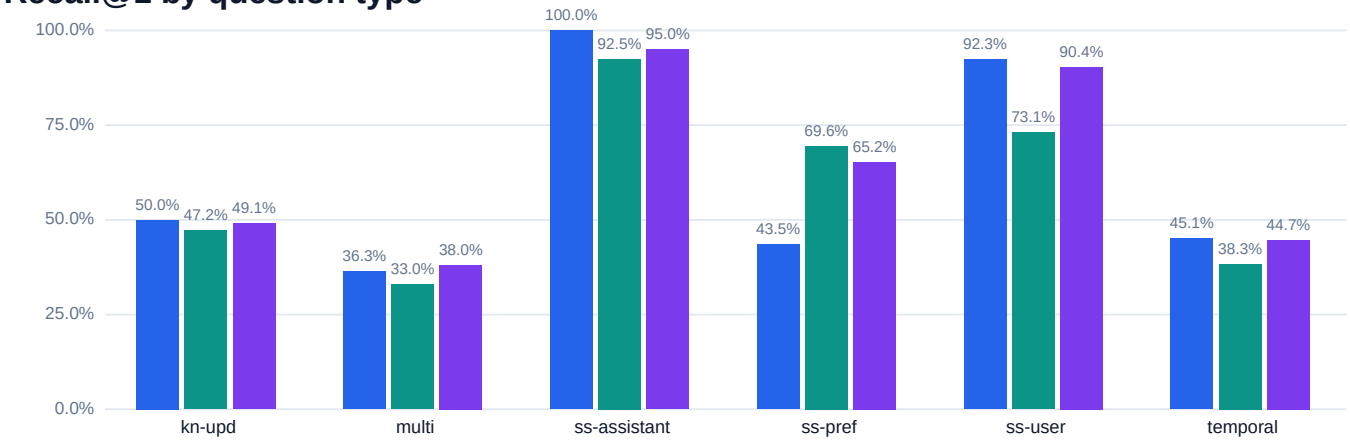
Configuration	R@1 (95% CI)	R@5	R@10	R@20 (95% CI)	NDCG@10	MRR@20	ms
v1 keyword	0.1% ±0.3	0.1%	0.1%	0.1% ±0.3	0.2%	0.3%	0.38
BM25	56.2% ±3.4	92.1%	95.2%	97.8% ±1.1	90.4%	91.6%	0.02
Vector (MiniLM)	51.1% ±3.6	90.7%	95.3%	98.3% ±1.0	87.0%	86.6%	0.04
Hybrid RRF	57.0% ±3.3	93.7%	96.9%	98.8% ±0.8	92.2%	93.0%	0.05
Hybrid + expansion	57.1% ±3.3	93.9%	97.1%	99.2% ±0.7	92.4%	93.1%	1.63

Recall@k curves



The curves flatten hard after 5. Almost everything is recoverable by rank 20, which is why Recall@20 cannot separate these systems; rank 1 is where the differences live.

Recall@1 by question type



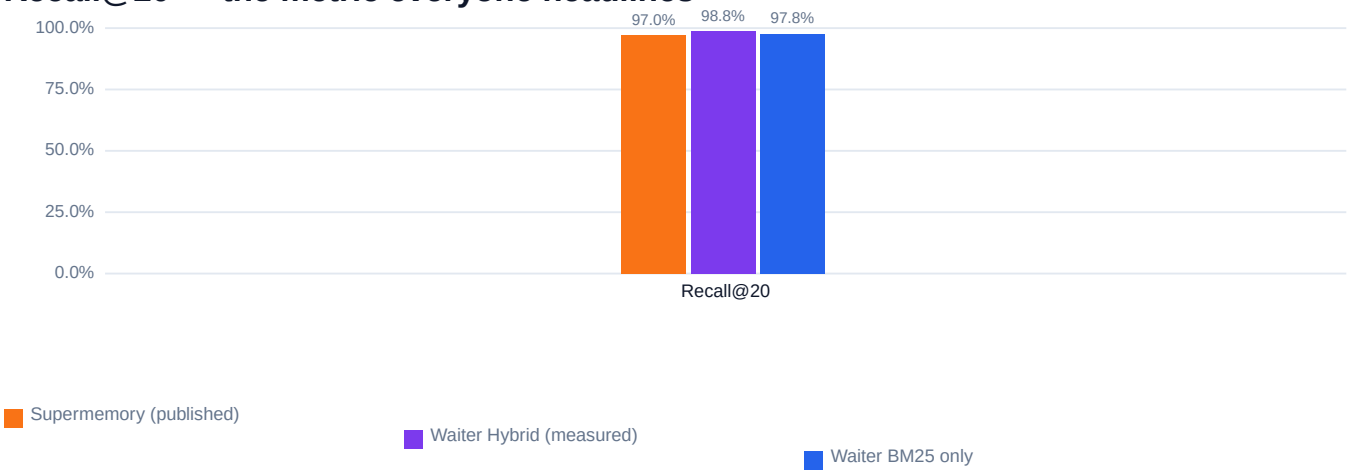
BM25 Vector Hybrid

Question type	n	BM25 R@1	Vector R@1	Hybrid R@1	Reads as
knowledge-update	54	50.0%	47.2%	49.1%	lexical: updated fact keeps its wording
multi-session	93	36.3%	33.0%	38.0%	fusion: needs several evidence sessions
ss-assistant	40	100.0%	92.5%	95.0%	lexical: digest misses assistant turns
ss-preference	23	43.5%	69.6%	65.2%	small n, indicative only · dense: implicit, no shared vocabulary
ss-user	52	92.3%	73.1%	90.4%	lexical: named entities match verbatim
temporal-reasoning	103	45.1%	38.3%	44.7%	weak for every retriever

3. Comparison with Supermemory and the published baselines

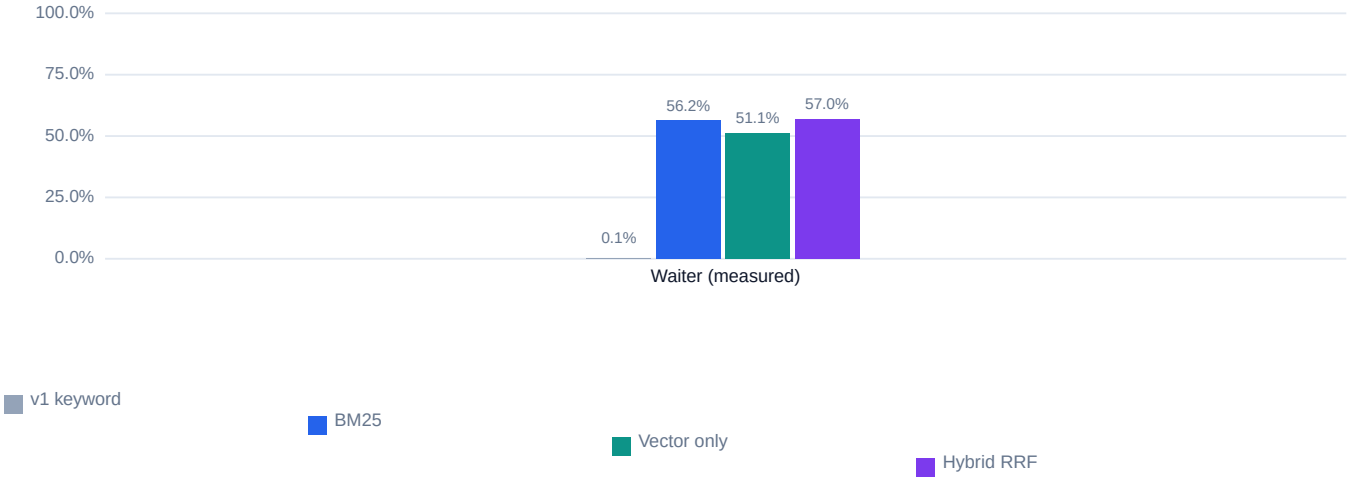
The published numbers and the numbers here come from different harnesses, so they are shown side by side, not merged. Supermemory reports Recall@20 with aggregation on LongMemEval-S; that is the one metric both can be measured on. Every other published figure is QA accuracy under an LLM judge, which this report does not produce.

Recall@20 — the metric everyone headlines



Waiter's Hybrid R@20 is 98.8% (99.2% with query expansion); Supermemory's vendor-reported result is 97.0%. This report has not reproduced Supermemory's aggregation/evaluation pipeline, so they are not a head-to-head win or parity result. Plain BM25 alone scores 97.8% on Waiter's split. With roughly 48 sessions per question, top-20 covers over 40% of the corpus, making this metric weakly discriminative here. Treat any cross-system difference as inconclusive without matched protocols and per-question results. See the public Supermemory report at supermemory.ai/research/longmembench/.

Where the retrievers actually differ



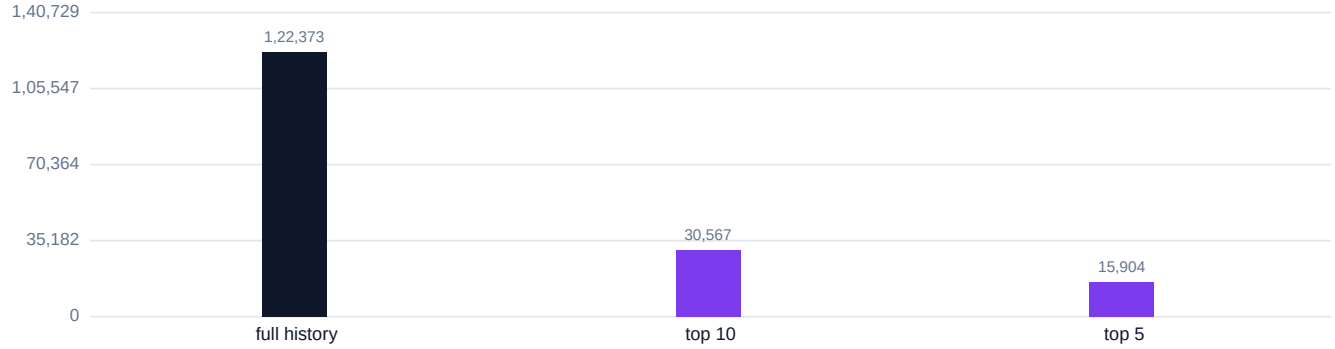
QA accuracy on LongMemEval-S (published, GPT-4o answerer and judge)

System	Overall accuracy	Source
Supermemory + GPT-4o	97.0%	supermemory research report
Supermemory + Gemini 3 Pro	85.2%	same report
Supermemory + GPT-5	84.6%	same report
Zep + GPT-4o	71.2%	same report
Full context + GPT-4o	60.2%	same report
Waiter	not measured	needs an answerer + judge with API access

Waiter has no row with a number in it on purpose. It is storage and retrieval with no LLM inside, so its answer accuracy is whatever model reads its output. Putting a figure there without running the harness would be fabrication.

4. Cost of answering a question

Tokens per question



Hybrid RRF finds an evidence session at rank 1 for 57.0% of questions; reading the top 5 costs 15,904 tokens instead of 1,22,373 — a 87% reduction in reading cost. That is the real product argument for retrieval here, more than the recall numbers.

Latency (warm, in-process)

Configuration	p50	Mean	Path
v1 keyword	0.38 ms	0.40 ms	substring scan over every memory, no ranking
BM25	0.02 ms	0.02 ms	inverted index, BM25 scoring (warm)
Vector (MiniLM)	0.04 ms	0.04 ms	brute-force cosine over the user's vectors
Hybrid RRF	0.05 ms	0.05 ms	BM25 + vector + RRF fusion
Hybrid + expansion	1.63 ms	1.65 ms	hybrid, run twice (expansion adds a second pass)

Brute-force cosine is fine at ~48 memories per user and will not be at 100k. An ANN index keyed by userId is the first thing to add before this is load-bearing. These timings exclude HTTP, ingestion, and embedding the query (~50 ms on this CPU).

5. What the numbers say

The v1 retriever was not retrieval

v1 scored 0.1% Recall@1 and 0.1% Recall@20. It searched for the literal question string inside a memory, which no conversation contains. In a demo with hand-written memories it looks fine; on real data it is dead. This is the most useful result in the report.

BM25 is the workhorse

Adding BM25 took Recall@1 from 0.1% to 56.2% and NDCG@10 to 90.4%, with no model, no GPU, and sub-millisecond queries. The questions and the evidence share names, products and places, and exact term matching is very good at that. Any plan that starts with an embedding pipeline and skips this step is spending money to lose accuracy.

Dense retrieval loses at rank 1, and wins exactly where you would expect

Vector-only scored 51.1% Recall@1 against 56.2% for BM25 — a 5.1 point deficit, roughly at the edge of the 95% intervals. The per-type table explains it: on single-session-assistant questions BM25 is perfect and the dense side drops 7 points, because the embedding only sees a digest of user turns and the evidence is in the assistant's reply. On single-session-preference questions the ranking flips: vector 69.6% against BM25 43.5%, because a preference is stated implicitly and shares no vocabulary with the question.

Fusion helps, but this run cannot prove it by much

Hybrid RRF reached 57.0% Recall@1 and 93.0% MRR@20; query expansion changes that to 57.1%. Against BM25 alone the gain is 0.7 points, well inside the ±3.3 confidence interval. Where it is visible is multi-session (38.0% vs 36.3% for BM25), which is the category that needs several evidence sessions. Reporting fusion as a big win would be overselling a real but small effect.

The recency prior was wrong, and tuning caught it

Down-weighting old sessions looked reasonable until it was measured: dev Recall@10 went 0!’98.4%, 0.5!’98.4%, 1!’97.3%, 2!’95.6%. The search selected 0, i.e. recency off. Most LongMemEval questions are not about the most recent thing, and a recency prior is a knowledge-update heuristic applied to everything. It stays off by default rather than on by intuition.

Query expansion is noise

Expanding the query with terms from the first pass moved Recall@1 by 0.1 percentage points and raised median latency from 0.05 ms to 1.63 ms (about 32× on this local microbenchmark). Pseudo-relevance feedback is only as good as the first pass, and when the first pass is already right there is nothing to recover.

The headline benchmark number is saturated

Recall@20 is 97.8% for BM25, 98.3% for vector, and 98.8% for hybrid. At ~48 sessions per question, the top 20 is 42% of the corpus. Vendors quoting improvements on this metric are mostly quoting the ceiling. If a Benchmark section is going to mean anything, it has to lead with Recall@1 or MRR, and it has to say which split and how many sessions per question.

Limitations

- QA accuracy was not measured. Nothing here supports a claim that Waiter answers as well as Supermemory.
- One embedding model (MiniLM-L6-v2) on one machine. bge-large or E5 would likely move the dense and hybrid numbers.
- Session digests, not chunk-level retrieval. A long session is matched on one 1.1k-character digest, which is exactly why assistant-turn questions suffer.
- No cross-encoder reranker. The reranking here is rank fusion, not a learned re-scorer.

- 'Hybrid' fuses only lexical and dense; no date-aware or entity-aware scoring is implemented.
- The cleaned split removed noisy sessions, making retrieval easier than the original LongMemEval-S.
- 365 test questions gives ±3.3 points at rank 1, so single-digit deltas need more data, not more opinion.

What to build next, in order

- Chunk-level memories with session-level aggregation: index each turn separately and score a session by its best passage. It directly targets the assistant-turn and multi-session deficits.
- A cross-encoder reranker over the fused top-30. Highest leverage quality step left, no schema change.
- An ANN index (HNSW) keyed by userId, before any of this is called at scale.
- Durable storage, so a 23k-session corpus does not vanish on restart.
- Run the QA half with a real answerer and judge, then put a labelled number in the accuracy table.

Appendix: reproducing this

Everything is in the repository; no API keys are involved at any step.

Step	Command
Download data	two HuggingFace URLs, see waiter/bench/README.md
Build embedding cache	npx tsx bench/embed.ts
Run the benchmark	npx tsx bench/run.ts
Render this PDF	npx tsx bench/report.ts

Configurations: v1 keyword, BM25, Vector (MiniLM), Hybrid RRF, Hybrid + expansion. Recency grid searched on dev at Recall@10: 0=98.4%, 0.5=98.4%, 1=97.3%, 2=95.6%; selected 0. Chat logs are full transcripts per session; the dense side embeds user turns only, capped at 1,100 characters.